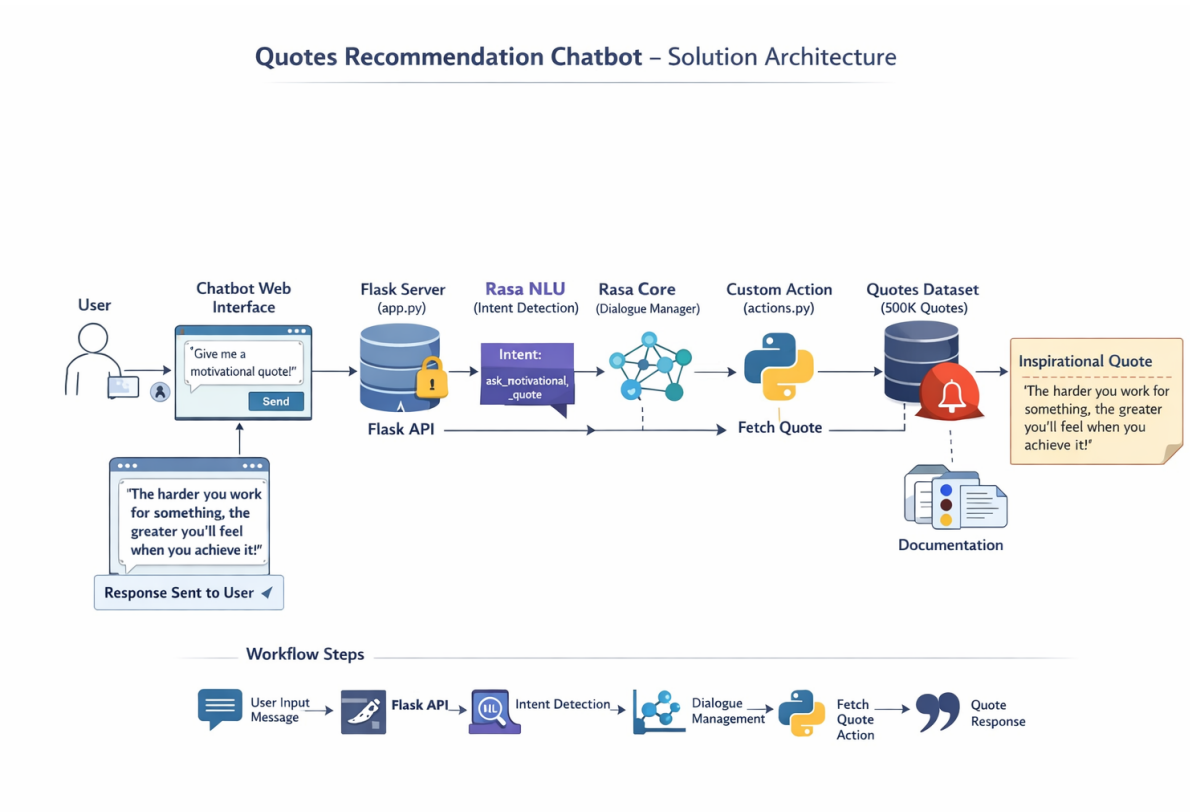
Project Design Phase-I
Solution Architecture

Date	5 March 2026
Team ID	Team 25
Project Name	Quotes Recommendation Chatbot using NLP
Maximum Marks	4 Marks

Solution Architecture:



Architecture Description

The **Quote Recommendation Chatbot using NLP** is designed using a layered architecture that connects user interaction with NLP processing and a quote database. **1. User Interface Layer**

- Provides the interface where users interact with the chatbot.
- Users enter **mood, keywords, or messages** to request quotes.
- Can be implemented using **HTML, CSS, JavaScript, or React**.

2. Application / Backend Layer

- Handles user requests and system logic.
- Manages **authentication, request handling, and communication** between components.
- Implemented using **Python with Flask or Django**.

3. NLP Processing Module

- Analyzes user input using **Natural Language Processing**.
- Performs:
  - Text preprocessing
  - Tokenization ○ Keyword extraction
  - Intent detection

4. Recommendation Engine

- Matches extracted keywords with quotes in the database.
- Uses **similarity algorithms or keyword matching** to select relevant quotes.

5. Database Layer • Stores: ○ Quotes ○ Authors ○ Categories ○ Keywords ○ User preferences

- Can use **MySQL, SQLite, or MongoDB**.

6. Response Generation

- Formats recommended quotes.
- Sends them back to the **chatbot interface for display to the user**.